

Ghada Almashaqbeh

Assistant Professor, University of Connecticut, Storrs, CT, USA

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Affiliated member at the Connecticut Advanced Computing Center (CACC), UConn

Affiliated member at the Engineering for Human Rights Initiative, UConn

RESEARCH INTERESTS

Cryptography, systems security, and privacy, with a primary focus on distributed cryptographic protocols, privacy-preserving computing, and blockchain-based systems.

EDUCATION

Columbia University	NY, USA	2015 - 2019
Ph.D. in Computer Science		2019
M.Phil in Computer Science		2018

- Thesis: *CacheCash: A Cryptocurrency-based Decentralized Content Delivery Network* (produced a startup—CacheCash Development Company, Inc.)
- Advisors: Allison Bishop and Tal Malkin

University of Notre Dame	IN, USA	2014 - 2015
Ph.D. in Computer Science (Transferred to Columbia)		

- Advisor: Marina Blanton

Jordan University of Science and Technology	Irbid, Jordan	2006 - 2008
M.Sc. in Computer Engineering		

- Thesis: *A Cross-Layer Based QoS Routing Framework for Wireless Mesh Networks*
- Advisors: Sameer Bataineh and Jamal Al-Karaki

The Hashemite University	Zarqa, Jordan	2001 - 2005
B.Sc. in Electrical and Computer Engineering		

EMPLOYMENT and PRIOR EXPERIENCES

University of Connecticut, CT, USA: *Assistant professor, School of Computing*, Aug 2020 - present

The Melon, MA, USA: *Advisor*, Apr 2022 - Mar 2023

Sunscreen, CA, USA: *Advisor*, Sep 2021 - Apr 2026

NuCypher, CA, USA: *Cryptographer*, Feb 2020 - Aug 2020, *Consultant*, Sep 2020 - Sep 2021

CacheCash Development Company, Inc., NY, USA: *Cofounder and Research Scientist*, Jun 2019 – Dec 2019

The Hashemite University, Zarqa, Jordan:

Lecturer in Computer Engineering, Feb 2014 - Jul 2014

Assistant Lecturer in Computer Engineering, Jun 2008 - Feb 2014

Lab Supervisor, Microprocessors Lab, Aug 2005 - Feb 2006

RESEARCH SUPPORT

Funded

- **DHS** (through the University of Nebraska)
Shield: Secure, Holistic Infrastructure for Election Logistics and Data
Total amount: **\$1,198,923** 9/1/2024 - 6/30/2026
Terminated on 3/17/2025 due to DHS funding priority changes
PI: Benjamin Fuller, Co-Is: Ghada Almashaqbeh (share 24%), Laurent Michel, Alexander Russell
- **Uniswap Foundation—TLDR Fellowship Program**
State Growth Control and Pricing for DeFi Systems
Total amount: **\$30,000** 10/15/2023 - 10/14/2024
PI: Ghada Almashaqbeh (share 100%)
- **Protocol Labs—Cryptonet Network Grant**
Timed, Anonymous and Revocable Delegation of Cryptographic Capabilities
Total amount: **\$34,999** 3/15/2023 - 3/15/2024
PI: Ghada Almashaqbeh (share 100%)
- **NSF Grant CNS-2226932 (SaTC: CORE Small)**
Towards Trustworthy and Performant Decentralized Resource Markets in the Blockchain Era
Total amount: **\$599,707** 10/1/2022 - 9/30/2027
PI: Ghada Almashaqbeh (share 100%)
- **UConn Research Excellence Program (REP) Award**
Anonymity and Restricted Interaction Patterns in Private Machine Learning
Total amount: **\$24,997** 6/1/2022 - 8/31/2023
PI: Ghada Almashaqbeh (share 100%)

In Submission

- NSF CAREER: *Security and Privacy of Heterogeneous Systems and Computing Models*, total amount: ~\$600,000, PI: Ghada Almashaqbeh

PUBLICATIONS

- In theory cryptography community, the convention is that authors are alphabetically ordered, while in security/systems community authors are ordered based on contribution. Papers with alphabetic author list start with **
- Wavy-underlined author names indicate students for whom I am the primary advisor
- Google Scholar link

Preprint / Under Review

1. **When Does Being Selfish Pay Off? Temporal Composability and Profitability in Selfish Mining**
C. Finkbeiner, C. Shaw, and G. Almashaqbeh
2. **SoK: Blockchain Oracle Services Between Theory and Practice**
C. Finkbeiner and G. Almashaqbeh
3. **chainScale: Secure Functionality-oriented Scalability for Decentralized Resource Markets**
M. Najd and G. Almashaqbeh

4. **MPC with Weighted and Fluid Participation**

******G. Almashaqbeh and S. Ghude

Refereed Conference Papers

5. **Parasol Compiler: Pushing the Boundaries of FHE Program Efficiency**

R. Weber, R. Orendorff, G. Almashaqbeh, and R. Solomon
S&P 2026

6. **Adversary Resilient Learned Bloom Filters**

******G. Almashaqbeh, A. Bishop, and H. Tirmazi
Asiacrypt 2025

7. **ammBoost: State Growth Control for AMMs**

N. Michel, M. Najd, and G. Almashaqbeh
DSN 2025

8. **SoK: Time to be Selfless?! Demystifying the Landscape of Selfish Mining Strategies and Models**

C. Finkbeiner, M. Najd, J. Guskind, and G. Almashaqbeh
ICBC 2025, also presented at SBC 2025

9. **Password-authenticated Cryptography from Consumable Tokens**

G. Almashaqbeh
CSCML 2024

10. **Anonymous, Timed and Revocable Proxy Signatures**

******G. Almashaqbeh and A. Nitulescu
ISC 2024 **Best Paper Award**

11. **Competitive Policies for Online Collateral Maintenance**

******G. Almashaqbeh, S. Chen, and A. Russell
AFT 2024

12. **chainBoost: A Secure Performance Booster for Blockchain-based Resource Markets**

Z. Motaqy, M. Najd, and G. Almashaqbeh
EuroS&P 2024

13. **Unclonable Cryptography: A Tale of Two No-cloning Paradigms**

******G. Almashaqbeh and R. Chatterjee
Secrypt 2023

14. **smartFHE: Privacy-Preserving Smart Contracts from Fully Homomorphic Encryption**

R. Solomon, R. Weber, and G. Almashaqbeh
EuroS&P 2023

15. **Unclonable Polymers and Their Cryptographic Applications**

******G. Almashaqbeh, R. Canetti, Y. Erlich, J. Gershoni, T. Malkin, I. Pe'er, A. Roitburd-Berman, and E. Tromer
Eurocrypt 2022

16. **SoK: Privacy-Preserving Computing in the Blockchain Era**

G. Almashaqbeh and R. Solomon
EuroS&P 2022

17. **Bet and Attack: Incentive Compatible Collaborative Attacks Using Smart Contracts**
Z. Motagy, G. Almashaqbeh, B. Bahrak, and N. Yazdani
 GameSec (Decision and Game Theory for Security), 2021
18. **MicroCash: Practical Concurrent Processing of Micropayments**
G. Almashaqbeh, A. Bishop, and J. Cappos
 FC (Financial Cryptography and Data Security), 2020
19. **CAPnet: A Defense Against Cache Accounting Attacks on Content Distribution Networks**
G. Almashaqbeh, K. Kelley, A. Bishop, and J. Cappos
 IEEE Conference on Communications and Network Security, 2019
20. **Public-Key Authentication for Cloud-based WBANs**
 T. Hayajneh, A. V. Vasilakos, G. Almashaqbeh, B. J Mohd, M. Shakir, K. Qaraqe and M. Imran
 The International Conference on Body Area Networks, 2014
21. **Multimedia traffic over WLANs: QoS support and performance evaluation**
 T. Hayajneh and G. Al-Mashaqbeh
 IEEE International Conference on Information and Communication Systems, 2014.
22. **Cloud-based Interference-aware Remote Health Monitoring System for Non-hospitalized Patients**
G. Almashaqbeh, T. Hayajneh, and A. V. Vasilakos
 IEEE Global Communications Conference, 2014
23. **Computers and e-Health: Roles and new applications**
G. Al-Mashaqbeh
 IEEE International Conference on Computer Systems and Industrial Informatics, 2012
24. **SENSORIA: A New Simulation Platform for Wireless Sensor Networks**
 J. Al-Karaki and G. Al-Mashaqbeh
 IEEE International Conference on Sensor Technologies and Applications, 2007

Refereed Journal Papers

25. **AnoFel: Supporting Anonymity for Privacy-Preserving Federated Learning**
G. Almashaqbeh and Z. Ghodsi
 Proceedings of Privacy Enhancing Technologies (PoPETs) Journal, 2025
26. **Gage MPC: Bypassing Residual Function Leakage for Non-Interactive MPC**
 **G. Almashaqbeh, F. Benhamouda, S. Han, D. Jaroslawicz, T. Malkin, A. Nicita, T. Rabin, A. Shah, and E. Tromer
 Proceedings of Privacy Enhancing Technologies (PoPETs) Journal, 2021
27. **Joint Geographic and Energy-aware Routing Protocol for Static and Mobile WSNs**
G. Al-Mashaqbeh, J. Al-Karaki, M. Al-Rousan, A. Raza, H. Abbas, and M. Pasha
 Ad hoc & Sensor Wireless Networks 41, 2018
28. **Implementing Support for Pointers to Private Data in a General-Purpose Secure Multi-Party Compiler**
 Y. Zhang, M. Blanton, and G. Almashaqbeh
 ACM Transactions on Privacy and Security, 21(2), 2017
29. **Routing Protocols in Wireless Mesh Networks: A Survey**
 J. Al-Karaki, G. Al-Mashaqbeh, and S. Bataineh
 International Journal of Information and Communication Technology 11, no. 4, 2017

30. **Secure Authentication for Remote Patient Monitoring with Wireless Medical Sensor Networks**
T. Hayajneh, B. Mohd, M. Imran, *G. Almashaqbeh*, and A. Vasilakos
Sensors 16, no. 4, 2016
31. **Secure Distributed Genome Analysis for GWAS and Sequence Comparison Computation**
Y. Zhang, M. Blanton, and *G. Almashaqbeh*
BMC Medical Informatics and Decision Making 15(Suppl 5), p. S4, 2015
32. **A Green Approach for Selfish Misbehavior Detection in 802.11-based Wireless Networks**
T. Hayajneh, *G. Almashaqbeh*, and S. Ullah
Mobile Networks and Applications, vol. 20, no. 5, 2015
33. **QoS-Aware Health Monitoring System Using Cloud-Based WBANs**
G. Almashaqbeh, T. Hayajneh, A. V. Vasilakos, and B. J. Mohd
Journal of Medical Systems, vol. 38, no. 10, 2014
34. **A Survey of Wireless Technologies Coexistence in WBAN: Analysis and Open Research Issues**
T. Hayajneh, *G. Almashaqbeh*, S. Ullah, and A. V. Vasilakos
Wireless Networks, vol. 20, no. 8, Springer, 2014
35. **An Energy-efficient and Security Aware Route Selection Protocol for Wireless Sensor Networks**
T. Hayajneh, R. Doomun, *G. Al-Mashaqbeh*, and B. J. Mohd
Security and Communication Networks, vol. 7, no. 11, 2014
36. **CLEAR: A Cross-layer Enhanced and Adaptive Routing Framework for Wireless Mesh Networks**
G. Al-Mashaqbeh, J. Al-Karaki, and S. Bataineh
Wireless Personal Communications, 2009
37. **Energy-Centric Routing in Wireless Sensor Networks**
J. Al-Karaki and *G. Al-Mashaqbeh*
Elsevier Microprocessors and Microsystems, vol. 31, no. 4, 2007

Refereed Workshop Papers

38. **ABC: A Cryptocurrency-Focused Threat Modeling Framework**
G. Almashaqbeh, A. Bishop, and J. Cappos
IEEE INFOCOM Workshops—Cryptocurrencies and Blockchains for Distributed Systems, 2019

Magazine Articles

39. **Rethinking Service Systems: A Path Towards Secure and Equitable Resource Markets**
G. Almashaqbeh
USENIX ;login: Magazine, 2021

HONORS AND AWARDS

- Best Paper Award—ISC 2024.
- TLDR Fellow—Uniswap Foundation, 2023.
- Foresight Institute Fellow, 2023.
- Research Award recognition (for awarded grants), Office of the Vice President for Research, UConn, 2022, 2023.
- UConn Research Excellence Award, 2022.

- Excellence in Teaching recognition, UConn, 2021.
- Grossman Scholar Award, Fu Foundation School of Engineering and Applied Science, Columbia University, 2018.
- CS PhD Service Award, Computer Science Department, Columbia University, 2018.
- Student travel grant, Crypto, 2018.
- CRA-W Grad Cohort Workshop scholarship, 2015, 2016, 2017.
- Grace Hopper Celebration of Women in Computing (GHC) student scholarship, 2016.
- Women in Theory Workshop scholarship, 2016.
- Indiana Celebration of Women in Computing Conference (InWIC) scholarship, 2015.
- First position in the 6th National Technological Parade, Jordan, 2013.
- Second position in the International IT Competition at Zayed University, UAE, 2011.
- Second position in the 4th National Technological Parade, Jordan, 2011.
- Ranked 1st among MSc students in the Computer Engineering Department, Jordan University of Science and Technology, Jordan, 2008.
- Ranked 1st among BSc students in the College of Engineering, and 2nd among all BSc students in the university, the Hashemite University, Jordan, 2005.
- Senior design project sponsorship award, King Abdullah II Design and Development Bureau (KADDB) and the King Abdullah II Fund for Development (KAFFD), Jordan, 2005.
- Dean's list, the Hashemite University, Jordan, 2001-2005.

TEACHING

Blockchain Technology.

- Spring 2026 - UConn CSE 3550/CSE 5000 Course Homepage
- Fall 2023 - UConn CSE 2550 Course Homepage
- Fall 2020, 2021, 2022 - UConn CSE 5095 Course Homepage
- Summer 2019 - Fordham University CISC 6880

Introduction to Computer and Network Security / Introduction to Cryptography and Cybersecurity

- Spring 2021, 2022, and 2023, Fall 2024 - UConn CSE 3400/CSE 5850 Course Homepage

Independent Study

- Graduate: Fall 2020, 2023, 2024, and 2025, Spring 2026 - UConn CSE 5099
- Undergraduate: Fall 2020, Spring 2021 - UConn CSE 4099

STUDENTS

Advising - PhD

- Mohamed Najd (Fall 2022 - present)
 - SYF (Synchrony Financial) Fellowship, Spring 2023
 - Thesis proposal done in Spring 2026, expected thesis defense/graduation in Spring 2027
- Colin Finkbeiner (Fall 2023 - present)
 - SYF (Synchrony Financial) Fellowship, Spring 2026

- Zahra (Raha) Motaqy (Fall 2021 - Spring 2023)
 - SYF (Synchrony Financial) Fellowship, Spring 2022 and Fall 2022
- Justin Yang (Fall 2022 - Fall 2023)
 - CSE Departmental Fellowship, academic year 2022/2023
- Sumedh Ghude (Fall 2024 - Fall 2025)

Advising - MSc

- Ronald Maule (graduated in Spring 2024), co-advised with Amir Herzberg
 - SYF (Synchrony Financial) Fellowship, academic year 2023/2024
- Nicolas Michel (graduated in Spring 2024)

Mentoring and collaboration

- Connor Shaw, undergrad (Spring 2026)
- Nicolas Michel, MSc (Fall 2023)
- Julia Guskind, undergrad (Spring 2023)
- Ronald Maule, undergrad (Summer 2022 and 2023)
- Allison Gagliano, MSc (Spring 2022)
- Bradshaw Pines, undergrad (Spring 2021)
- Pablo Rodriguez, undergrad (academic year 2020/2021)—McNair Scholar

Committees

- PhD Qualifying Exam committee: Naeem Ahmed, Maryam Kafikang, Fariha Boishakhi, Yifan Zhang, Md Mahbub Hasan, Amey Shukla, Shuai Shao, Dingbang Wang
- PhD Thesis committee: Nidhibahen Shah, Justin Furuness, Sara Wrotniak, Chloe Cachet

PROFESSIONAL SERVICE

- Mentor at the IEEE Computer Society's Speed Mentorship session - IEEE S&P conference, 2026.
- NSF panels (two panels) - 2025.
- Grant proposal reviewer, the Foresight Institute, 2025 - present.
- Mentor for junior PC members - IEEE EuroS&P conference, 2024 and 2025
- Web 3.0 startups mentor at Frequency Crypto Bootcamp (hosted by Pillar VC and The Melon), Oct 2022
- Early career mentor - IEEE EuroS&P conference, 2022 and 2024
- **Service at UConn:**
 - Mentor for the S-STEM Program at the College of Engineering, Spring 2025 - present.
 - The brown bag faculty-student lunch, Fall 2024 (discuss CSE major concentrations and undergrad research opportunities).
 - Workload committee, 2024/2025, 2025/2026
 - Qualifying Exam committee, Spring 2024, 2024/2025
 - Performance Review committee, 2023/2024
 - Reviewer for the UConn Research Excellence Award (REP), Spring 2023
 - MS DS concentration advisor—Cybersecurity, 2022/2023, 2023/2024, 2024/2025
 - Blockchain/Fintech Majors Curriculum Task Force, Summer and Fall 2022
 - Doctoral Awards committee, Spring 2022
 - Judge for the 8th Annual UConn School of Engineering Poster Competition, Spring 2022
 - Diversity and Inclusion committee, Fall 2021 - Spring 2026
 - Graduate Curriculum Consilium, Fall 2021 - Spring 2026
 - Co-organizer of the CSE Security Seminar, 2020 - 2022

- Faculty Search committee, 2020/2021
- Strategic Planning committee, 2020/2021
- Judge for SDP (senior design project), Spring 2021
- **Service at Columbia:**
 - Ph.D coordinator of the Emerging Scholars Program (ESP) at Columbia University, 2017-2018
- **Organizing committee member:**
 - S&P’26 and ’27 (Workshops Chair).
 - S&P’25 (DEI Chair, and Student Travel Grant committee member).
- **Program committee member:**
 - *Conferences*: Eurocrypt’27, S&P’27, CCS’26, Usenix Security’26, S&P’26, D2’26, FHE.org’26, FC’26, CCS’25, FC’25, PETS’25, EuroS&P’25, AsiaCCS’25, ISC’24, CCS’24, PETS’24, EuroS&P’24, FC’24, Crypto’23, CCS’23, FC’23, EuroS&P’23, PKC’23, Asiacrypt’22, Crypto’21.
 - *Workshops*: NESD’24, WPES’22, CFAIL’19 and ’20.
 - Applied Research Competition - NYU Cyber Security Awareness Week (CSAW’16 and ’17).
- **Reviewer (other than what is listed under PC above):**
 - *Conferences*: CCS’22, EuroS&P’22, Eurocrypt’21, Eurocrypt’20, TCC’18, USENIX Security’18, USENIX ATC’17, Eurocrypt’17, CCS’16.
 - *Workshops*: WPES’22.
 - *Journals*: ACM Transactions on Privacy and Security, Information Processing Letter, IEEE/ACM Transactions on Networking, Journal of Human Rights, Springer Wireless Networks, IEEE Systems, Wireless Personal Communication, Journal of Medical Systems, Pervasive and Mobile Computing.

TALKS AND PANELS

Talks

- *Anonymity in Decentralized Settings*
 - Anonymity Day Workshop, Columbia University - Apr 2025
- *Count Me In, or Not?! Crowdsourcing in the Age of Blockchains*
 - New York Celebration of Women in Computing (NYCWIC) - Apr 2025
- *Password-authenticated Cryptography from Consumable Tokens*
 - CSCML’24 - Dec 2024
- *Anonymous, Timed and Revocable Proxy Signatures*
 - ISC’24 - Oct 2024
- *Give and Take; The Evolving Relationship between Security and Blockchains*
 - Northeastern University, Khoury College of Computer Science Security Seminar - Oct 2024
- *chainBoost: A Secure Performance Booster for Blockchain-based Resource Markets*
 - EuroS&P’24 - Jul 2024
- *ammBoost: State Growth Control for AMMs*
 - The TLDR Conference - May 2024
- *smartFHE: Privacy-Preserving Smart Contracts from Fully Homomorphic Encryption*

- EuroS&P’23 - Jul 2023
- *On the Power of Smart Contracts—The Good and The Bad*
 - Heritage Medical Systems annual meeting - Dec 2022
 - Software Center/MDU Workshop on Cybersecurity - Sep 2022
- *Unclonable Polymers and Their Cryptographic Applications*
 - QSig Workshop, University of Edinburgh - Jan 2024
 - Intelligent Cooperation Group Seminar, Foresight Institute - Aug 2023
 - New York Celebration of Women in Computing (NYCWIC) - Apr 2023
 - Software Systems and Cybersecurity Seminars, Monash University - Mar 2023
 - NY Crypto Day - Oct 2022
 - UConn CSE Security Seminar - Oct 2022
 - Foresight Institute Crypto, Security & AI Workshop - Oct 2022
 - CMU CyLab Crypto Seminar - Jul 2022
 - Eurocrypt’22 - Jun 2022
- *Cybersecurity and Society—Blockchain Technology*
 - Orientation to Engineering Course OCP, UConn - Nov 2022
 - The UConn Explore Engineering (E²) Program - Jun 2022
- *I See You Blockchain User, or Not! Privacy in the Age of Blockchains*
 - USENIX Enigma’22 - Feb 2022
- *Gage MPC: Bypassing Residual Function Leakage for Non-Interactive MPC*
 - Protocol Labs Research Seminar Series - Feb 2022
 - PETS’21 - Jul 2021
 - Google Cryptography Talks Series - Jul 2021
 - UConn CSE Security Seminar - Jul 2021
- *Rethinking Service Systems: A Path Towards Secure and Equitable Resource Markets*
 - Grace Hopper Celebration (GHC), Security/Privacy track - Oct 2020
- *Micropayments: From Centralized to Blockchain-based Distributed Schemes*
 - University of Malaga - May 2020
- *CAPnet: A Defense Against Cache Accounting Attacks on Content Distribution Networks*
 - IEEE CNS’19 - Jun 2019
- *CacheCash: A Cryptocurrency-based Decentralized Content Delivery Network*
 - PhD dissertation defense, Columbia University - May 2019
- *The Age of Cryptocurrencies: Bitcoin and Sisters*
 - University of Colorado Colorado Springs - Mar 2018 and Apr 2019
 - NYU Tandon School of Engineering - Dec 2017
 - Columbia University - Dec 2017
- *Threat Modeling for Cryptocurrency-based Systems*
 - NYU Tandon School of Engineering - Dec 2018
- *Resource-backed Cryptocurrencies*
 - Association of Women in Math (AWM) Talk Series, Barnard College - Nov 2018.

- Emerging Scholars Program Seminar, Columbia University - Dec 2017
- *Sensible Cryptocurrencies*
 - PhD Candidacy Exam Talk, Columbia University - Nov 2017
- *Cryptocurrency Era*
 - Fordham University - Jun 2017
- *Bitcoin*
 - NYU Tandon School of Engineering - Dec 2015
- *Digital Currencies*
 - Cybersecurity for STEM Teachers Summer Program, NYU Polytechnic School of Engineering - Jul 2015

Panels

- *Challenges Up the Ladder: Funding, Networking and Industry Support*, N2Women event at the IEEE LCN conference - Sep 2022
- *Exploring the Interplay of Privacy, Security, and Blockchain Technologies*, Women in Security and Privacy (WISP) and Blockchain Lawyers Forum (BLF) - Nov 2021
- *Blockchain Coffee Talk*, Synchrony Financial (SYF) - May 2021
- *Crypto-Economics 101*, the 6th Annual Entrepreneurship Festival StartupColumbia, Columbia University - Apr 2019